

# Iniyen Andrews Joseph

Doctoral Applicant in Computer Science | Aalto University & ELLIS Institute Finland

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Location: Espoo / Helsinki Region, Finland | Primary Call: Autumn 2026 ELLIS Doctoral Programme

## RESEARCH INTERESTS

**Stochastic Control & Deep Reinforcement Learning (PPO)**, Generative World Models & Diffusion Processes, High-Dimensional Non-Stationary State-Space Modeling, 3D Kinematics & Biomechanical Digital Twins, and Multimodal Neuro-Symbolic Foundation Architectures.

## EDUCATION

**National Institute of Electronics & Information Technology (NIELIT)**, Calicut, India

*Master of Science in Computer Science (Specialization: Data Science)*

**Expected May 2026**

**CGPA: 8.55 / 10.0**

- ▶ **M.Sc. Thesis:** *Autonomous High-Frequency Trading via Deep Reinforcement Learning (Project Bithax)*
- ▶ **Supervisor:** Dr. Vimala Mathew (Scientist E, Department of IT, National Institute of Electronics & Information Technology (NIELIT), Calicut)
- ▶ **Methodology & Focus:** Formulated continuous high-frequency market dynamics as a continuous state-space Markov Decision Process (MDP) over a 42-dimensional state vector. Engineered PPO with volatility-penalized variance terms and transaction friction constraints, eliminating look-ahead bias via walk-forward out-of-sample validation.

**The American College**, Madurai, India

*Bachelor of Science in Computer Science*

**2021 – 2024**

**CGPA: 7.4 / 10.0**

## RESEARCH & TECHNICAL PROJECTS

**Procedural Kinematic Synthesizer (30 Articulated Joints)**

**2025 – 2026**

*Lead Architect & Researcher* | *Three.js, WebGL, Analytical Kinematics (72-Hr Engine Sprint)* | [\[Live Demo\]](#) | [\[Proposal PDF\]](#)

- ▶ **Articulated Kinematic Core & 5 Lexicons:** Engineered an in-browser, 60 FPS Kinematic Synthesizer (30 articulated joints) parameterized across 5 sub-lexical spaces (41 handshapes, 24 loci, 10 orientations, 16 movements, 4 bilateral modes) generating 629,760 discrete phonetic configurations.
- ▶ **Universal Generator & ASL-LEX Coverage:** Built a universal parametric sign compiler spanning the complete 2,723-sign ASL-LEX 2.0 vocabulary without manual rigging, complete with OOV procedural token synthesis.
- ▶ **Zero Inter-Finger Self-Collision Kinematics:** Engineered 7-point phalanx soft-body repulsion barriers, joint-space anti-penetration solvers, and dorsal clamps; automated headless audit verifies **minimal finger collisions** across all 26 alphabet letters and benchmark words.
- ▶ **Battison's Laws & Quartic Strain Optimization:** Operationalized Battison's Symmetry and Dominance laws with real-time quartic strain minimization  $P_{\text{strain}}(\theta)$  and closed-form 2-bone arm IK with continuous volumetric obstacle avoidance.

**Continuous Trajectory Planning via Deep Reinforcement Learning (Trackmania)**

**2024 – 2025**

*Lead Researcher* | *Python, PyTorch, Gymnasium, Stable-Baselines3, PPO*

[\[View on GitHub: Joeytribb\]](#)

- ▶ **Continuous Action-Space Control:** Formulated high-speed vehicle trajectory planning as a continuous action-space Markov Decision Process (MDP). Engineered policy gradient algorithms (PPO) with dynamic reward shaping to navigate non-linear contact friction, high-speed inertial drift, and latency-bounded state estimation.
- ▶ **Stochastic Policy Generalization:** Investigated policy convergence and sample efficiency under noisy perceptual feedback, evaluating temporal state representations to stabilize control across non-stationary track environments.

RESEARCH & TECHNICAL PROJECTS (CONTINUED)

**Generative World Models for High-Frequency Stochastic Microstructure** 2026  
Lead Researcher | Python, PyTorch, Transformers, Generative Modeling [View on GitHub: Generative\_LOB\_Simulator]

- **Autoregressive Microstructure Modeling:** Developed autoregressive generative sequence architectures to model the high-dimensional, asynchronous event dynamics of Limit Order Books (LOB).
- **Non-Markovian Order Flow Simulation:** Investigated transformer-based world models to capture non-Markovian queue depletion, order-flow transitions, and fat-tailed price impact distributions in financial microstructure, establishing a computational framework for generative simulation.

**Variance Reduction in High-Noise Regimes: Deep Learning vs. Regularized Baselines (Project HOPE)** 2024 – 2025  
Independent Researcher | Scikit-Learn, TensorFlow, LSTM Networks [View on GitHub: HOPE]

- **Low-SNR Generalization Benchmark:** Conducted a systematic empirical study comparing deep recurrent neural networks (LSTMs) against regularized linear state estimators for daily directional return prediction across the S&P 500 and Nifty 50 indices.
- **Variance-Constrained Optimization:** Demonstrated that in low signal-to-noise ratio (SNR) stochastic regimes, strongly regularized linear models outperform over-parameterized deep neural networks by constraining estimator variance, attaining an annualized Sharpe Ratio > 2.5 and 70.6% win rate.

**Jaguu: Multimodal Edge Foundation Agent** 2025  
Lead Developer | Llama 3, Ollama, PyTorch, Computer Vision [View on GitHub: Jaguu]

- **Local Foundation Agent Architecture:** Architected a privacy-first, local multimodal voice and reasoning agent coupling quantized autoregressive LLMs (Llama 3) with computer vision backbones.
- **Sub-Second Edge Token Inference:** Engineered low-latency cross-modal token conditioning pipelines for compute-constrained edge deployment, achieving sub-second token response on consumer-grade hardware.

TECHNICAL & MATHEMATICAL COMPETENCIES

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|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stochastic Control & RL:       | Markov Decision Processes (MDPs), Proximal Policy Optimization (PPO), Deep Q-Networks (DQN), Risk-Sensitive Reward Engineering, Stable-Baselines3, Gymnasium.                             |
| 3D Kinematics & Graphics:      | Cyclic Coordinate Descent (CCD) IK, Analytical 2-Bone Arm IK, Lie Algebra ( $\mathfrak{se}(3)$ ), Quaternions, Euler Transforms, Three.js, WebGL, Biomechanical Hand/Arm Rigs, Blender3D. |
| Deep Learning & Foundation AI: | PyTorch, Hugging Face Transformers, Autoregressive LLMs (Llama 3), Diffusion Models, Fully Sharded Data Parallel (FSDP), WebDataset, Distributed Training.                                |
| Systems & HPC Infrastructure:  | Python (Advanced), C++, JavaScript/ES6, SQL, Linux, Git, Docker, $\LaTeX$ , AMD ROCm / HIP Environments (CSC LUMI Supercomputer Readiness).                                               |

SELECTED GRADUATE COURSEWORK & ACADEMIC MERIT

**Relevant Graduate Coursework (National Institute of Electronics & Information Technology (NIELIT), Calicut):** Advanced Machine Learning, Deep Learning Architectures, Reinforcement Learning, Stochastic Processes & Probability, Statistical Data Modeling, Advanced Algorithms, Data Structures in C++.

**Academic Standing:** First Class with Distinction, M.Sc. Computer Science (CGPA: 8.55 / 10.0).

ACADEMIC REFERENCES

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| <b>Dr. Marco Raoul Marini</b><br>Research Mentor (SLT/SLP Formalization)<br>Sapienza University of Rome<br>Rome, Italy<br>Email: <a href="mailto:marcoraoul.marini@uniroma1.it">marcoraoul.marini@uniroma1.it</a> | <b>Dr. Vimala Mathew</b><br>Scientist E & M.Sc. Thesis Supervisor<br>Department of Information Technology<br>National Institute of Electronics & Information Technology (NIELIT)<br>Calicut, Kerala, India<br>Email: <a href="mailto:vimala@calicut.nielit.in">vimala@calicut.nielit.in</a> |
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